

SightPath — AI-Assisted Wearable Navigation for the Visually Impaired

Theme: Create technology-driven solutions addressing real-world challenges faced by communities across India — applying software, AI, IoT, and data science to improve accessibility, sustainability, inclusion, and quality of life.

DOMAIN

Accessibility & Assistive Tech

CATEGORY

HARDWARE

SOFTWARE

PS TYPE

Open Innovation

BACKGROUND

India is home to over 4.9 crore people with visual impairment — one of the largest such populations in the world — yet most Indian cities lack tactile paving, audio signals, or consistent infrastructure for safe independent mobility. Existing aids such as the white cane or guide dogs help detect nearby obstacles but cannot warn users of hazards at head height, moving vehicles, potholes, or unmarked construction — challenges that are especially common on Indian streets. This forces many visually impaired individuals to depend on a companion for even routine outdoor tasks, limiting their independence, employability, and quality of life.

PROBLEM STATEMENT

Design and build an affordable, AI-powered wearable and companion mobile application that helps visually impaired users navigate Indian streets, public transport, and indoor spaces safely and independently — detecting obstacles in real time, reading signage and text aloud, and guiding the user with simple audio or haptic cues, without relying on expensive or import-dependent hardware.

EXPECTED FOCUS AREAS

- Real-time obstacle & hazard detection using low-cost sensors (ultrasonic / depth camera)
- Voice- and haptic-based turn-by-turn guidance, usable without constant internet access
- Emergency SOS alert to family or nearby volunteers with live location
- On-device computer vision for text, signage, and face-familiarity recognition
- GPS + IoT integration for public transport stop and route identification
- Offline-first, low-power design suited for everyday, all-day wearability

EXPECTED OUTCOME

A working prototype (wearable/handheld device + companion app) that demonstrably detects obstacles and hazards in real time, converts visual information such as text and signboards into speech, and guides a visually impaired user along a short indoor or outdoor route with measurable improvement in walking confidence and reduced dependence on a sighted companion.

Solutions will be evaluated on real-world usability, affordability of the underlying design, offline reliability, response latency, and inclusiveness of design — including support for regional languages and low-literacy users.

IMPACT

ACCESSIBILITY

Enables independent, safer mobility for millions of visually impaired citizens across India.

INCLUSION

Improves access to education, employment, and public services for persons with disabilities.

SCALABILITY

Low-cost, open hardware design allows deployment in both urban and rural communities.